International Risk-Sharing in a Fragmented World

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The views expressed herein are those of the authors and not necessarily those of the Federal Reserve Bank of Minneapolis or the Federal Reserve System.

Motivation

- The world economy is increasingly fractured along geopolitical lines
  - Economic relations are shaped not only by economic fundamentals, but also by political alliances and strategic rivalries

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  - Does financial fragmentation hinder international risk-sharing?

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  - How does geopolitical risk shape the international allocation of capital?
  - Does financial fragmentation hinder international risk-sharing?
- This paper: **fragmentation in official bilateral lending** — facts and theory.

What we do

Fragmentation of official government-to-government lending

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- Financial fragmentation index

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2. Geopolitically aligned countries have more synchronized business cycles
3. Fragmentation weakens lending flows from low-risk to high-risk countries

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Theory of fragmentation:

- Geopolitical rivalry ⇒ creditor identity matters for repayment incentives
- Rising geopolitical risk ⇒ endogenous fragmentation

Related Literature

- **Geoeconomics:** Aiyar, Malacrino, and Presbitero (2024); Aiyar, Presbitero, and Ruta (2023); An and Huber (2026); Sturm Becko and O'Connor (2024); Bianchi and Sosa-Padilla (2024, 2025); Broner, Martin, Meyer, and Trebesch (2024); Caldara and Iacoviello (2022); Catalán, Fendoglu, and Tsuruga (2024); Clayton, Coppola, Maggiori, and Schreger (2024, 2025); Clayton, Maggiori, and Schreger (2024, 2025); De Souza, Hu, Li, and Mei (2024); Fernandez-Villaverde, Mineyama, and Song (2024); Gopinath, Gourinchas, Presbitero, and Topalova (2024); Kempf, Luo, Schäfer, and Tsoutsoura (2023); Kleinman, Liu, and Redding (2024); Pflueger and Yared (2024); Sturm Becko (2024)

  Fragmentation of official lending: facts and theory

- **Sovereign debt and limited commitment:** Aguiar & Amador (2011); Aguiar & Gopinath (2006); Arellano (2008); Broner, Martin & Ventura (2010); Eaton & Gersovitz (1981)

  Geopolitical risk and implications for capital flows and risk-sharing

Data

Bilateral government-to-government lending (Horn, Reinhart & Trebesch, 2026)

- Loans, grants, and guarantees by governments and state-controlled agencies
- Lending flows between 120 creditor and 190 debtor countries, 1910–2024
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Official lending is an ideal laboratory for studying geoeconomic fragmentation

- Dyadic data: creditor identity is directly observed
- Long-run coverage spans major geopolitical episodes
- State control makes geopolitics first-order

Measuring fragmentation

Goal: measure whether official lending follows geopolitical alignment

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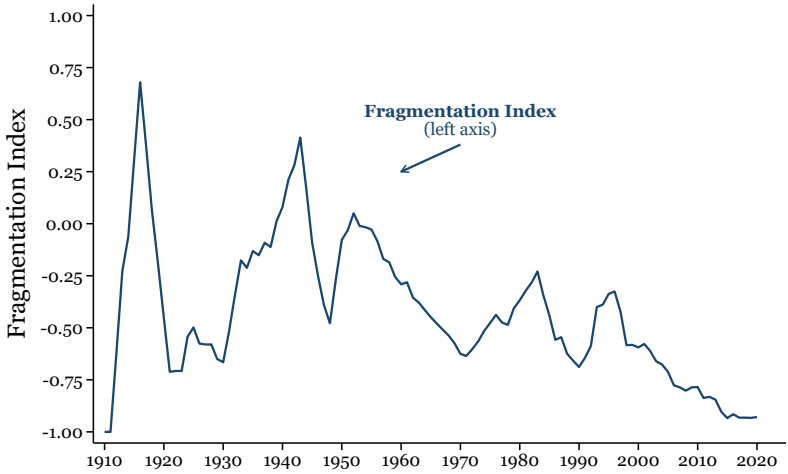
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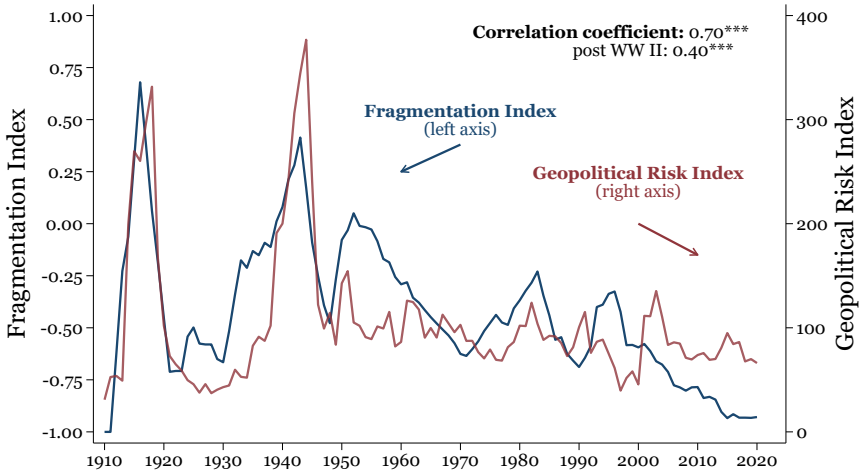
Identifying Allies and Non-Allies:

- Military alliances as coded by the Correlates of War Project (Gibler & Sarkees 2004)
- Robustness: UN General Assembly voting similarity (Bailey et al. 2017)

Fragmentation and geopolitical risk, 1910–2022

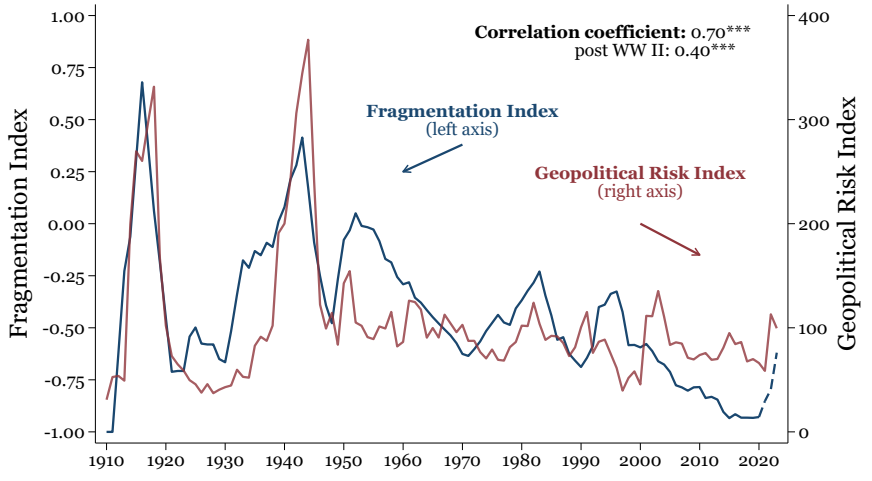


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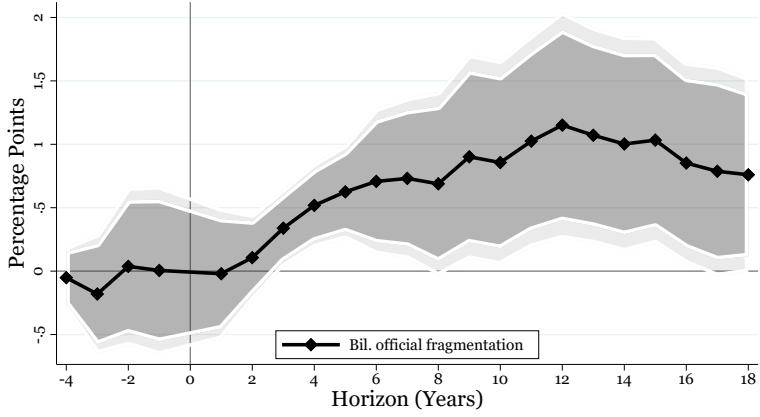
FINDING 1: A rise in geopolitical risk leads to higher financial fragmentation

Geopolitical risk and borrowing fragmentation

$$F_{i,t+h} - F_{i,t-1} = \alpha^h + \beta^h \log \mathrm{GPR}_{i,t} + \sum_{\ell=1}^{4} \gamma_\ell^h F_{i,t-\ell} + \mathrm{controls}_{i,t} + \epsilon_{i,t}^h$$

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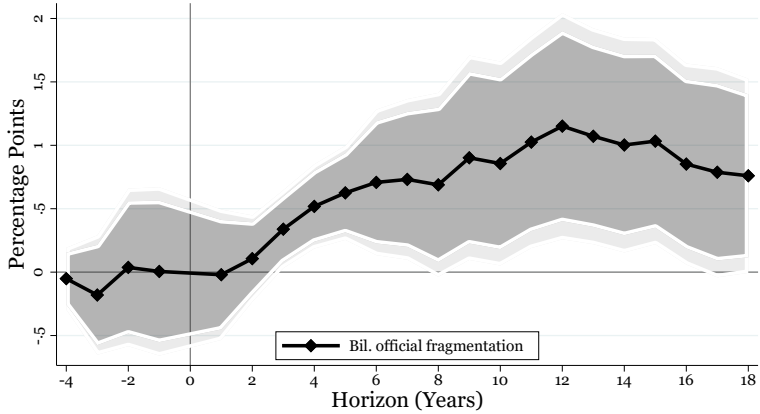
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A 10% ↑ in geopol. risk persistently increases fragmentation. 1.2pp (at peak)

Controls: GDP, trade, tail-risk and crises indicators. [+]

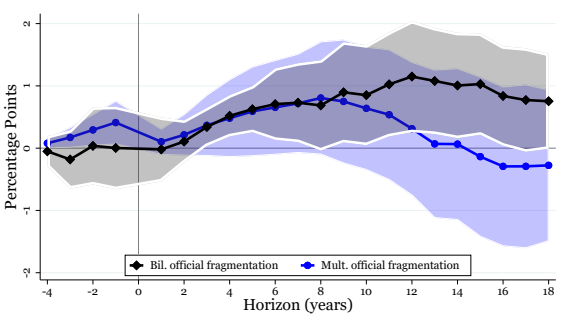
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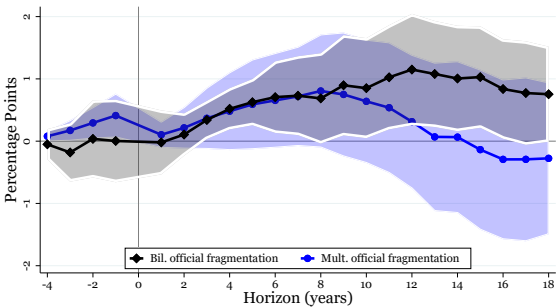


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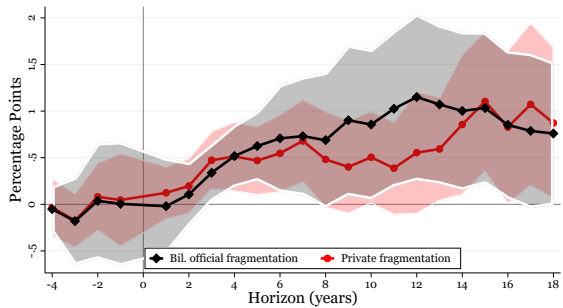
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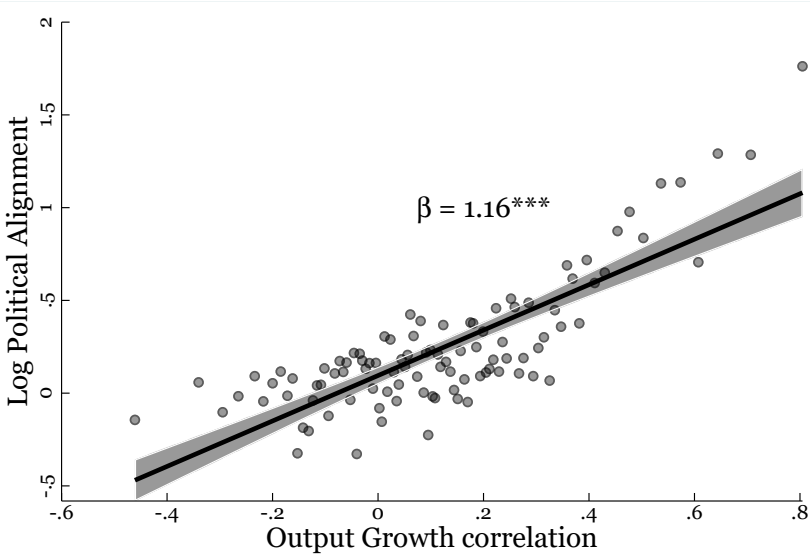
Bilateral vs. private lending, 1970–2024



Little systematic multilateral response; smaller but significant private response. [+]

FINDING 2: Aligned countries have synchronized business cycles

Geopolitical alignment and output-growth correlations



▸ Tail risk

FINDING 3: Lending within alliances provides weaker insurance

Fragmentation and Risk-Sharing

Gravity specification: official credit flows, tail-risk gaps, and political alliance

$$\text{Flow}_{ijt} = \exp\Big[\beta\left(\text{TailRisk}_{i,t}^{debtor} - \text{TailRisk}_{j,t}^{creditor}\right) + \gamma\,\text{Alliance}_{ijt} + \psi\,\text{Alliance}_{ijt} \times \left(\text{TailRisk}_{i,t}^{debtor} - \text{TailRisk}_{j,t}^{creditor}\right) + \theta_i + \delta_j\Big]\epsilon_{ijt}$$

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- $\beta > 0$: official credit flows from lower-risk creditors to higher-risk borrowers

  $\Rightarrow$ official lending provides insurance

- $\gamma > 0$: allied countries lend more to each other

- $\psi < 0$: the response to tail-risk differences is weaker inside alliances

  $\Rightarrow$ within-bloc lending provides weaker insurance

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Tail-risk difference	3.840***	3.788***	4.222***
	(0.388)	(0.405)	(0.435)
Alliance		0.406***	0.427***
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Tail-risk difference × Alliance			−2.206*
			(1.162)
Observations	60,108	60,108	60,108
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Taking stock

1. Higher geopolitical risk is associated with greater fragmentation in official lending
2. Aligned countries have more synchronized output growth and tail risk
3. Lending within alliances provides weaker insurance against tail risk

Higher geopolitical risk is associated with more concentrated lending among allies, where the scope for risk-sharing is smaller

Model

Model Environment

- Two periods, one aggregate shock, three countries (home, allied, and rival)

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- **Period 1:** Home signs bilateral state-contingent contracts
  - Home: risk-averse. Allied and Rival: risk-neutral
- **Period 2:** Aggregate shock $s \in \{s_g, s_b\}$ with $y(s_b) < y(s_g)$
  - Home repays or defaults

Geopolitical Externality

Home country's objective at $t = 1$:

Geopolitical externality

$$\sum_s \pi(s) u\big(c(s)\big) \; - \; \eta \, V^*$$

where V^* is the Rival's expected payoff.

- $\eta > 0$ captures the desire to gain strategic advantage over geopolitical rivals

Repayment and Default

Repayment, $d(s) = 0$:

$$c(s) = y(s) - \tilde{a}(s) - a^*(s)$$

Payments To Ally

Payments To Rival

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Payments To Ally — Payments To Rival

Default, $d(s) = 1$:

Keep assets if positive

$$c(s) = (1 - \phi)\, y(s) - \tilde{a}_-(s) - a^*_-(s), \qquad x_- \equiv \min\{0, x\}$$

Income cost of default

- **Non-discriminatory default:** Home cannot selectively default on Rival alone

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  - Empirical support ▸ Arrears

Ally and Rival Countries

- Period-1 utility values is given by expected consumption

$$\tilde{V}(\tilde{a}, s) = \sum_{s \in S} \pi(s) \left[\tilde{y}(s) + (1 - d(s))\tilde{a}(s) + d(s)\tilde{a}_{-}(s)\right]$$

$$V^*(a^*, s) = \sum_{s \in S} \pi(s) \left[y^*(s) + (1 - d(s))a^*(s) + d(s)a^*_{-}(s)\right]$$

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- Consumption must be positive in each state

$$\tilde{y}(s) + (1 - d(s))\tilde{a}(s) + d(s)\tilde{a}_{-}(s) \geq 0$$

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The Incentive Constraint

Geopolitical cost of repaying Rival

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When $\eta = 0$: IC depends only on total repayment $\tilde{a} + a^*$

When $\eta > 0$: defaulting becomes more tempting the larger is the share of debt owed to rival

Optimal Bilateral Contracts

$$\max_{c(s),\,\tilde{a}(s),\,a^*(s)} \sum_s \pi(s) u(c(s)) \;-\; \eta V^*$$

subject to

Participation: $\displaystyle\sum_s \pi(s)\,\tilde{a}(s) \geq 0, \quad \sum_s \pi(s)\,a^*(s) \geq 0$

Incentives: $u(c(s)) - \eta(y^*(s) + a^*(s)) \geq u\big((1-\phi)y(s) - a^*_-(s) - \tilde{a}_-(s)\big)$

$$-\eta\big(y^*(s) + a^*_-(s)\big) \quad \forall s$$

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Incentive constraint binds in the good state because govt. repays in good state

$\Rightarrow$ limits insurance in the bad state

Simplifying Assumptions

- **Equal state probabilities:**

$$\pi(s_g) = \pi(s_b) = 0.5 \quad \Rightarrow \quad \tilde{a}(s_g) = -\tilde{a}(s_b) \quad \text{and} \quad a^*(s_g) = -a^*(s_b)$$

  - Good-state repayment equals the bad-state insurance transfer.

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- **Log utility:**

$$u(c) = \log(c)$$

Proposition, part i: Full Insurance

If $\phi \geq \hat{\phi}$, we have **full insurance**:

$$c(s_g) = c(s_b) = \frac{y(s_g) + y(s_b)}{2}$$

$$\tilde{a}(s_g) + a^*(s_g) = \frac{y(s_g) - y(s_b)}{2}$$

Thresholds: $\hat{\phi} \equiv 1 - \frac{y(s_g)+y(s_b)}{2y(s_g)} \exp\left(-\eta \left[\frac{y(s_g)-y(s_b)}{2} - \tilde{y}(s_b)\right]\right)$ $\underline{\phi} \equiv \frac{\tilde{y}(s_b)}{y(s_g)}$

Proposition, part ii: Imperfect Insurance

If $\phi < \hat{\phi}$, we have **imperfect insurance**:

$$c(s_g) > c(s_b)$$

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$$c(s_g) > c(s_b)$$

Imperfect insurance takes two forms:

- Partial fragmentation, $\underline{\phi} < \phi < \hat{\phi}$: Ally maxes out

$$\tilde{a}(s_g) = \tilde{y}(s_b), \qquad a^*(s_g) < \frac{y(s_g) - y(s_b)}{2} - \tilde{y}(s_b)$$

Thresholds: $\hat{\phi} \equiv 1 - \frac{y(s_g)+y(s_b)}{2y(s_g)} \exp\left(-\eta \left[\frac{y(s_g)-y(s_b)}{2} - \tilde{y}(s_b)\right]\right)$ $\qquad \underline{\phi} \equiv \frac{\tilde{y}(s_b)}{y(s_g)}$

Proposition, part ii: Imperfect Insurance

If $\phi < \hat{\phi}$, we have **imperfect insurance**:

$$c(s_g) > c(s_b)$$

Imperfect insurance takes two forms:

- Partial fragmentation, $\underline{\phi} < \phi < \hat{\phi}$: Ally maxes out

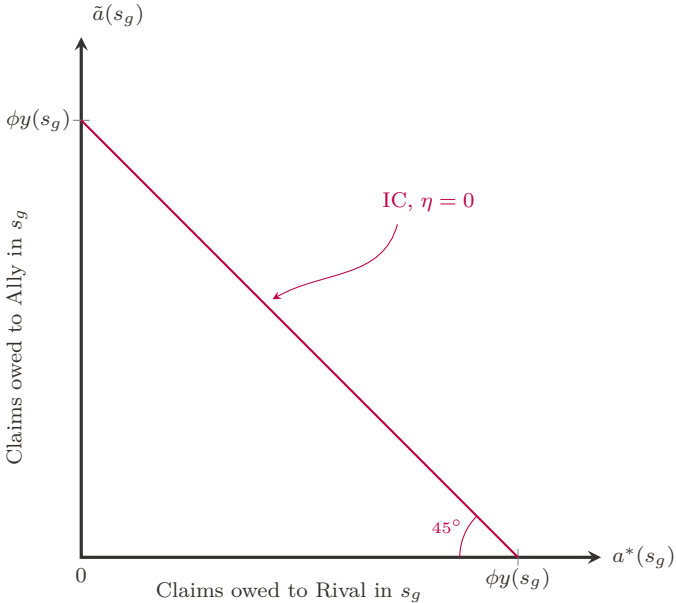
$$\tilde{a}(s_g) = \tilde{y}(s_b), \qquad a^*(s_g) < \frac{y(s_g) - y(s_b)}{2} - \tilde{y}(s_b)$$

- Complete fragmentation, $\phi \leq \underline{\phi}$: No borrowing from rival

$$\tilde{a}(s_g) = \phi y(s_g) \leq \tilde{y}(s_b), \qquad a^*(s_g) = 0$$

Thresholds: $\hat{\phi} \equiv 1 - \frac{y(s_g)+y(s_b)}{2y(s_g)} \exp\left(-\eta \left[\frac{y(s_g)-y(s_b)}{2} - \tilde{y}(s_b)\right]\right)$ $\qquad \underline{\phi} \equiv \frac{\tilde{y}(s_b)}{y(s_g)}$

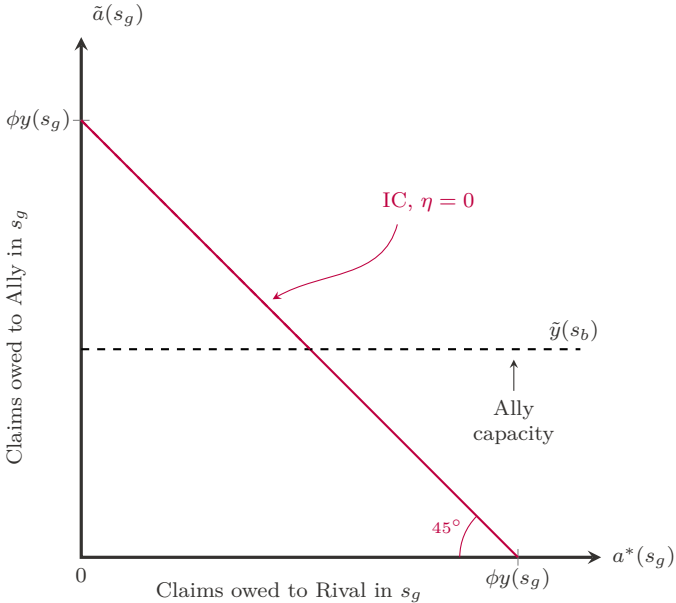
How Geopolitical Risk leads to Fragmentation



Benchmark: $\eta = 0$

Only **total debt** matters: split between Ally and Rival is irrelevant

How Geopolitical Risk leads to Fragmentation

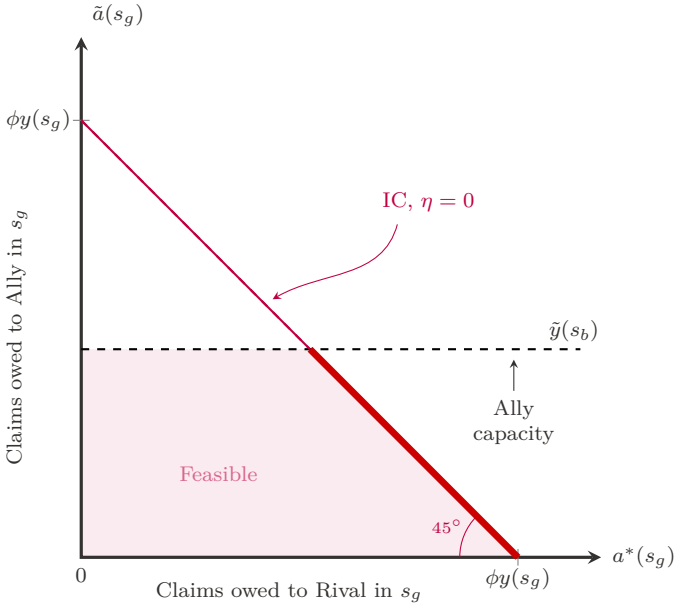


Ally capacity

The Ally cannot fully insure Home on its own.

So full insurance requires some lending from the Rival.

How Geopolitical Risk leads to Fragmentation

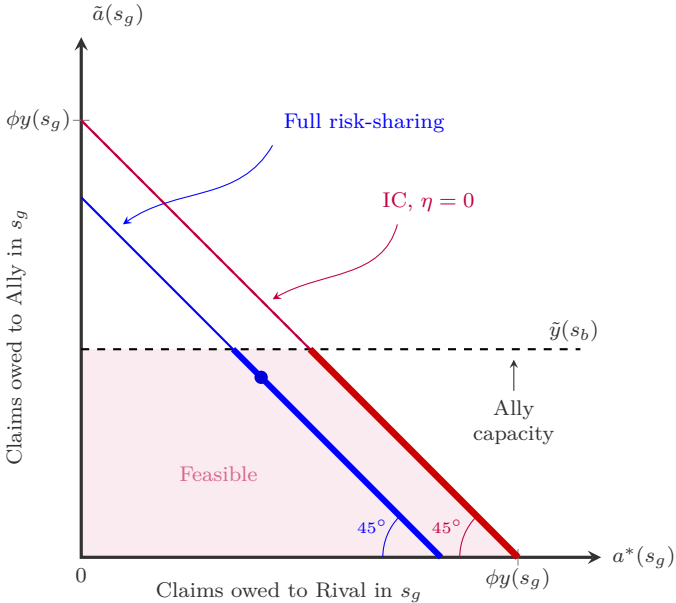


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How Geopolitical Risk leads to Fragmentation

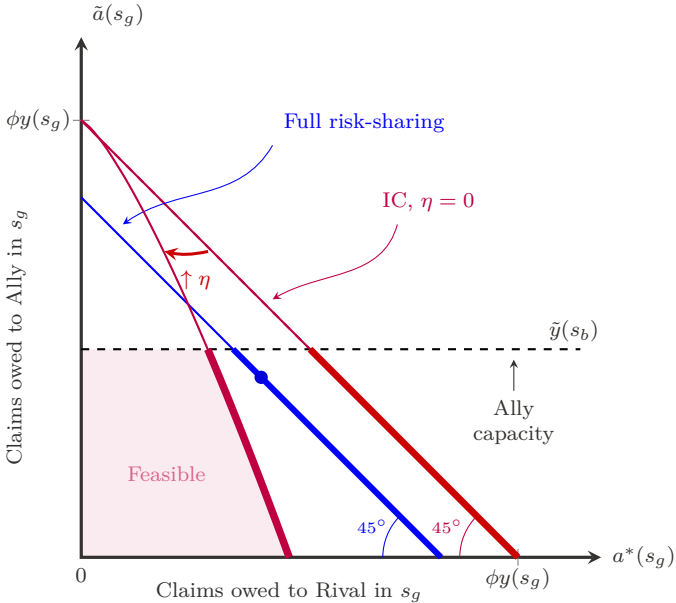


Full insurance is feasible

Home uses both creditors.

Consumption can be fully smoothed across states.

How Geopolitical Risk leads to Fragmentation

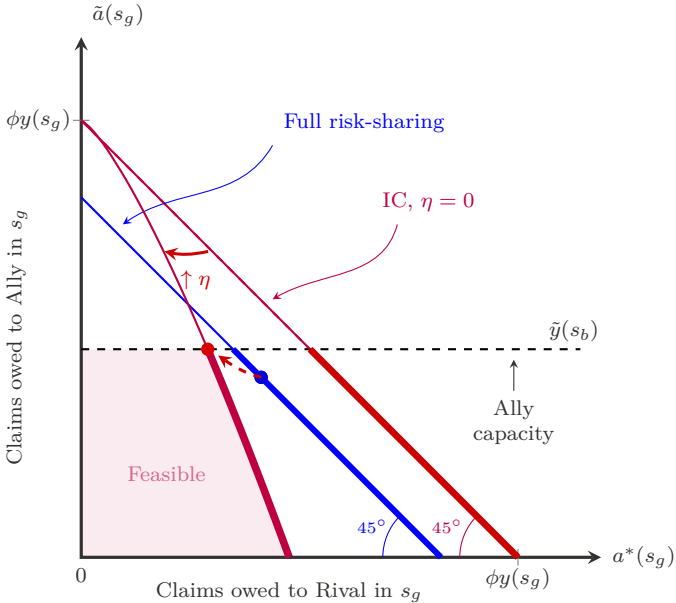


Increase in η

Repaying Rival becomes more costly.

More borrowing from Ally

How Geopolitical Risk leads to Fragmentation

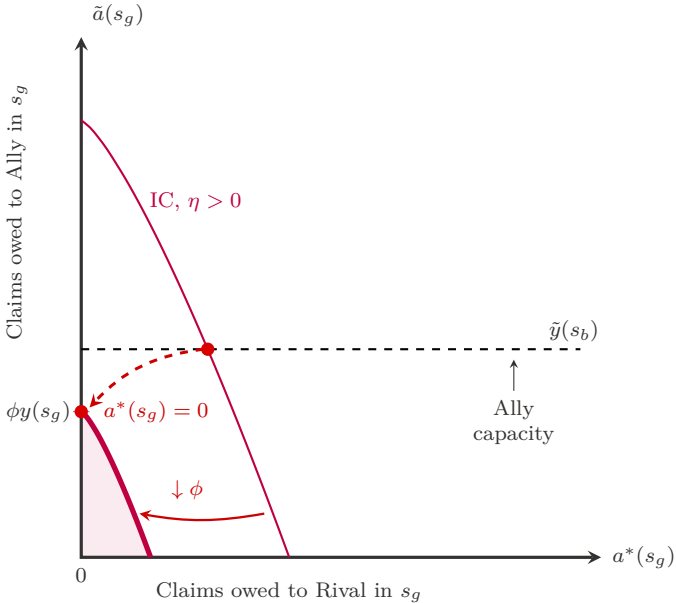


Full insurance breaks down

The Ally maxes out.

Rival lending falls.

How Geopolitical Risk leads to Fragmentation



Complete fragmentation

If the default cost is sufficiently low, Home borrows only from the Ally.

$$a^*(s_g) = 0.$$

Conclusions

- We use a long-run dyadic dataset on government-to-government lending to document and examine financial fragmentation
- Higher geopolitical risk is associated with more concentrated lending among allies, where the scope for risk-sharing is smaller
- We build a sovereign debt model with limited commitment that links strategic rivalry to fragmentation in official lending and limits to risk-sharing

Appendix

Data on private lending

World Bank International Debt Statistics (IDS) and Graf von Luckner and Horn (2026)

- Long-term private lending to public and publicly guaranteed borrowers, 1970–2024
- USD 1.69 trillion (constant 2020 USD); 116 creditor and 115 debtor countries
- Commercial bank loans, bonds, supplier credit and other private credit

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Coverage limits

- Creditor identity is measured on a residency basis
- Offshore intermediation can obscure the ultimate creditor
- Secondary-market trading of bonds is not captured

Back to LPs

Data on multilateral lending flows

Back to LPs

- In Bianchi et al. (2026), we combine data on multilateral lending with a granular new dataset on the funding structures of international financial institutions

Data on multilateral lending flows

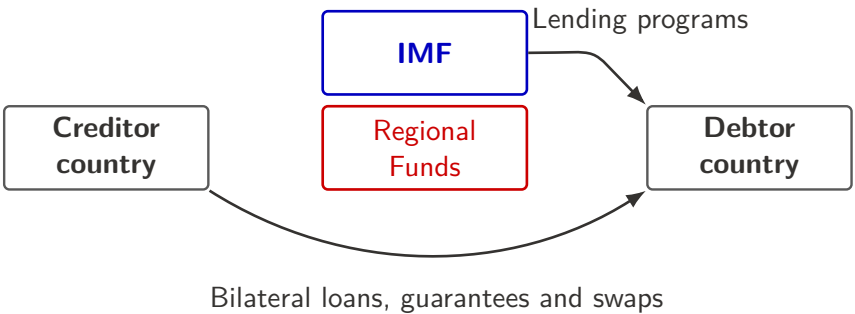
Back to LPs

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Data on multilateral lending flows

Back to LPs

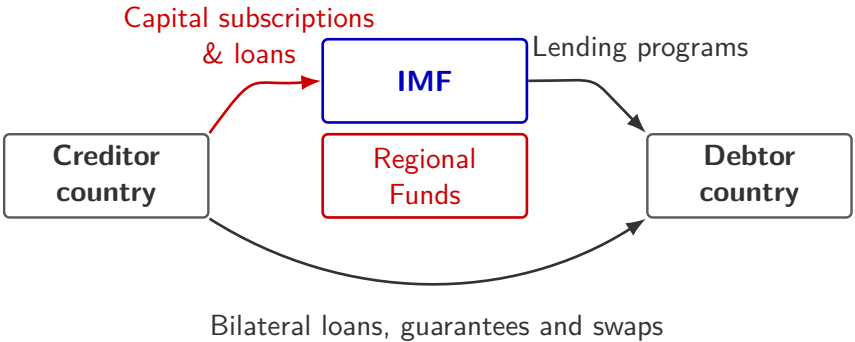
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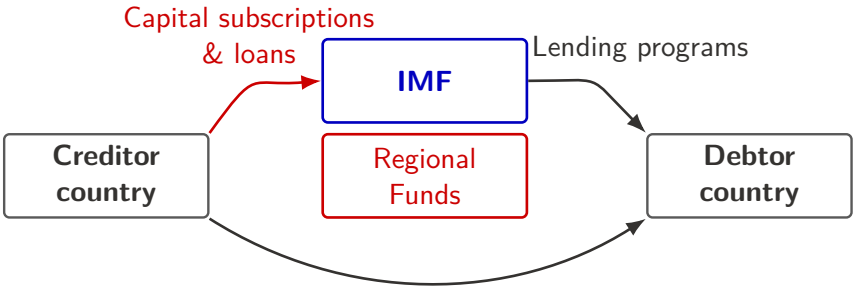
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Related approach:
Coppola et al. (2021)

Imputing lending through multilateral institutions

1920–2024: 16 institutions, 130 creditor and 190 debtor countries

- We compute the funding share of creditor country j in organization o in period t

$$\omega_{jot} = \frac{PAID.IN_{jot} + CREDIT_{jot}}{\sum_{k=1}^{N}(PAID.IN_{kot} + CREDIT_{kot})}$$

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- Total imputed lending: USD 2.3 trillion in constant 2020 USD. Aggregate across institutions:

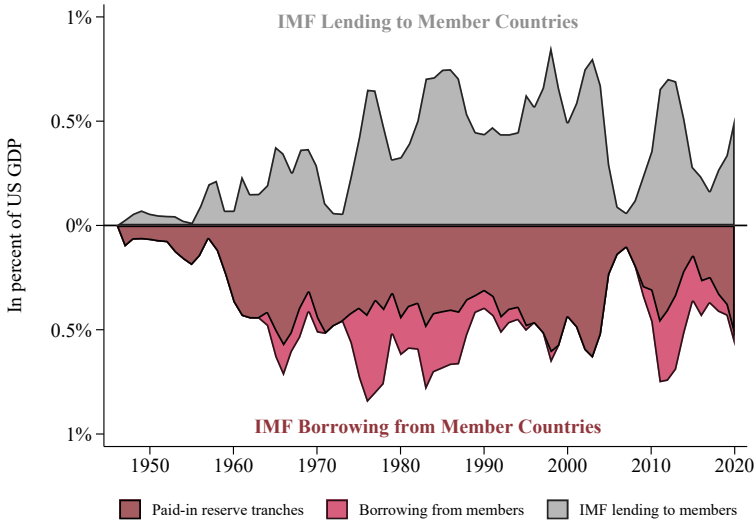
$$\sum_{o} LOAN_{ijot}$$

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IMF borrowing and lending

Historical funding illustration, through 2020

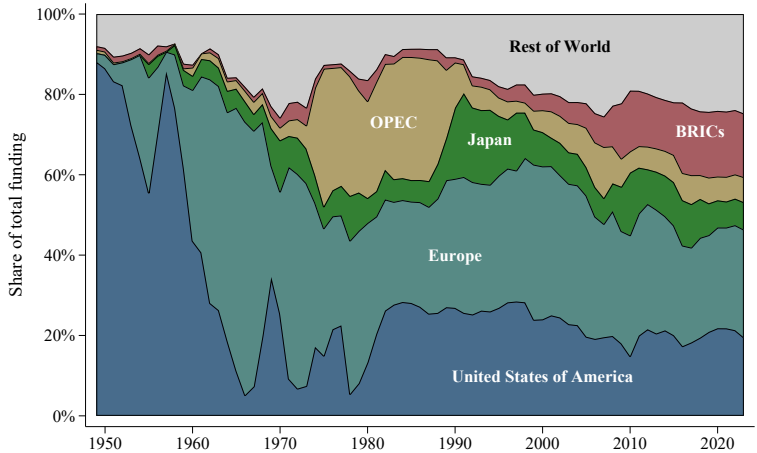
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Example: Who funds the IMF?

Historical funding illustration, through 2020

Back to LPs



Sources

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Agreement to establish Andean Reserve Fund, 1976

Capital

Article 5. The initial capital of the Fund is five hundred million ($500,000,000) dollars of the United States of America, suscribed as follows:

Bolivia:	sixty-two million five hundred thousand ($62,500,000) dollars.
Colombia:	one hundred twenty-five million ($125,000,000) dollars.
Ecuador:	sixty-two million five hundred thousand ($62,500,000) dollars.
Peru:	one hundred twenty-five million ($125,000,000) dollars.
Venezuela:	one hundred twenty-five million ($125,000,000) dollars.

Agreement to establish Euopean Monetary Fund, 1955

CONTRACTING PARTIES	AMOUNT OF CONTRIBUTIONS *(in units of account)*
Germany	42,000,000
Austria	5,000,000
B.L.E.U.	30,000,000
Denmark	15,000,000
France	42,000,000
Greece	2,850,000
Iceland	1,000,000
Italy	15,000,000
Norway	15,000,000
Netherlands	30,000,000
Portugal	5,000,000
United Kingdom	86,575,000
Sweden	15,000,000
Switzerland	21,000,000
Turkey	3,000,000
TOTAL	328,425,000

Multilateral institutions in the paper sample

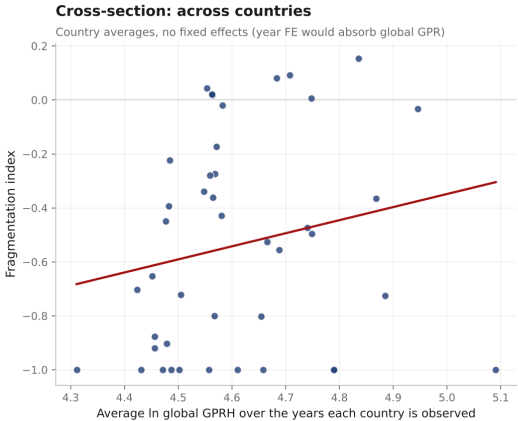
Back to LPs

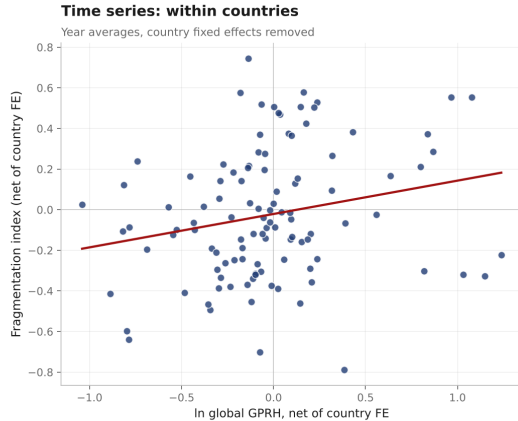
Capital and membership figures as reported in the June 2026 paper

Institution	Operating time	Authorized capital (in bn USD)	Number of member countries
League of Nations	1920 - 1946	n.a.	63
International Monetary Fund	1946 - 2024	1350	189
Andean Reserve Fund	1978 - 1991	2	5
Arab Monetary Fund	1977 - 2024	5	22
BRICS Contingent Reserve Arrangement	2014 - 2024	100	5
Chiang Mai Initiative	2000 - 2024	240	10
Eurasian Anti-Crisis Fund	2009 - 2024	9	6
European Monetary Fund	1958 - 1973	0.6	16
European Community Loan Mechanism	1975 - 1988	n.a.	12
European Financial Assistance Facility	1975 - 1988	n.a.	12
European BOP Facility	1988 - 2024	60	28
European Financial Stability Facility	2010 - 2013	1040	19
European Financial Stability Mechanism	2010 - 2013	75	28
European Stability Mechanism	2012 - 2024	780	19
Latin American Reserve Fund	1991 - 2024	4	8
NAFTA Swap Facility	1994 - 2024	7	3

GPR and fragmentation within and across countries

back

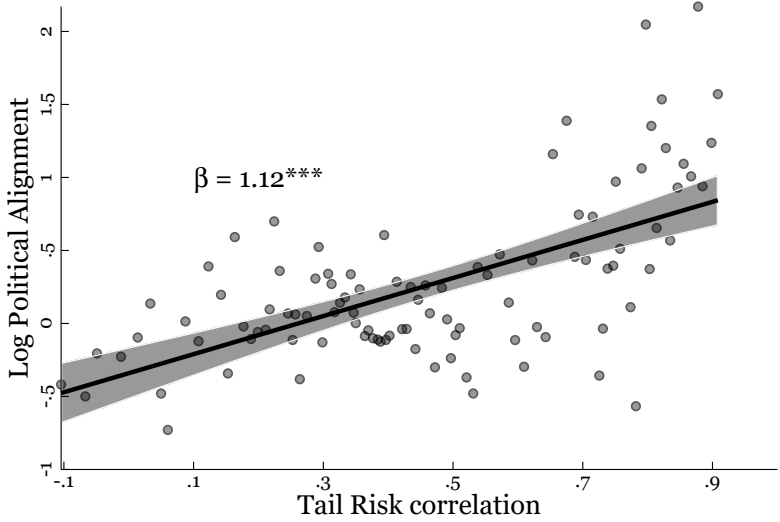




Baseline fragmentation index. x-axis: Caldara–Iacoviello global historical GPR (GPRH), in logs. Global GPR is identical across countries within a year, so the cross-section x-value only reflects which years a country is in the sample. Sample: 2,041 country-years, 1900–2024.

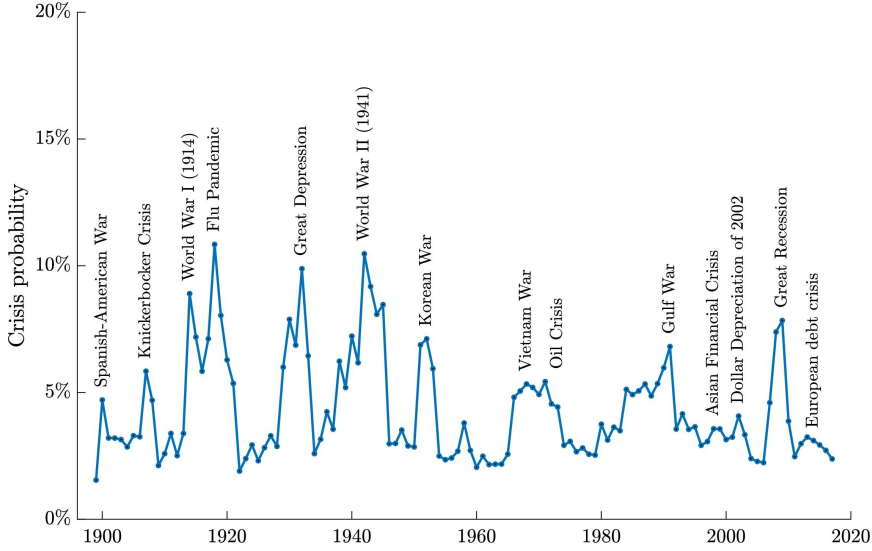
Geopolitical alignment and tail-risk correlations

back



Macroeconomic tail risk, 1900 - 2020, Aggregate

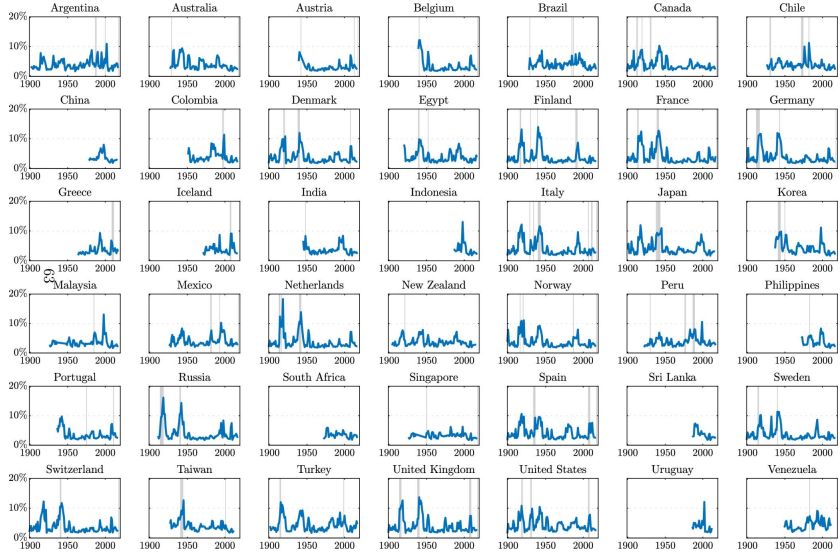
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Source: Marfe & Penasse (JFE, 2024)

Macroeconomic tail risk, 1900 - 2020

back



Source: Marfe & Penasse (JFE, 2024)

Payment arrears on allies and rivals

back

High-geopolitical-risk default episodes, 1970–2020

